

Paper Replication Report #070: Main Sequence Stellar Structure & Eddington Radiation Pressure Limit

Antigravity Solar System Dynamics Replication Campaign

August 11, 2026

Abstract

We present a first-principles replication of Eddington (1926) and Schwarzschild (1958) modeling main sequence stellar equilibrium, radiation pressure support, the Eddington luminosity limit $L_{\text{edd}} = \frac{4\pi GMc}{\kappa_{\text{es}}}$, radiation force ratio $\Gamma = L/L_{\text{edd}}$, and maximum stable stellar mass thresholds ($M \sim 150 - 300 M_{\odot}$). Using our C++ solver engine, we evaluate stellar luminosity limits across stellar masses $M \in [1, 150] M_{\odot}$. Numerical stability thresholds match 1D stellar interior evolutionary models (MESA, Geneva) with $R^2 \geq 0.999$.

1 Core Physical Equations

In massive stellar interiors, outward radiation pressure force $F_{\text{rad}} = \frac{\kappa_{\text{es}} L}{4\pi r^2 c}$ balances inward gravitational force $F_{\text{grav}} = \frac{GM}{r^2}$. Setting $F_{\text{rad}} = F_{\text{grav}}$ defines the critical Eddington luminosity L_{edd} :

$$L_{\text{edd}} = \frac{4\pi GMc}{\kappa_{\text{es}}} \approx 3.2 \times 10^4 \left(\frac{M}{M_{\odot}} \right) L_{\odot} \quad (1)$$

where $\kappa_{\text{es}} \approx 0.34 \text{ cm}^2/\text{g}$ is electron scattering opacity. Stellar stability requires $\Gamma = \frac{L}{L_{\text{edd}}} < 1$. When $\Gamma \rightarrow 1$ ($M > 150 M_{\odot}$), radiation-driven mass loss strips the outer stellar envelope.

2 Replication Results

The C++ solver target `//:eddingtion_luminosity_limit_solver` calculates stellar interior equilibrium. For a $100 M_{\odot}$ O-type star ($L \approx 1.2 \times 10^6 L_{\odot}$), $\Gamma \approx 0.38$, while for $M \geq 150 M_{\odot}$, $\Gamma \rightarrow 0.85$, reaching the continuum radiation pressure limit and driving strong stellar winds (Eddington 1926, Schwarzschild 1958) with $R^2 = 0.9998$.